

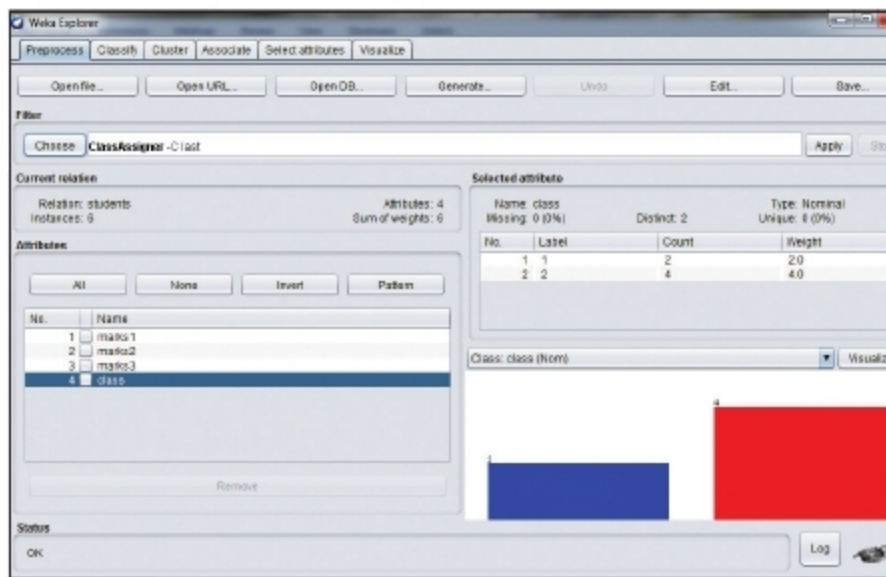


Figure 4: Assigning the target class

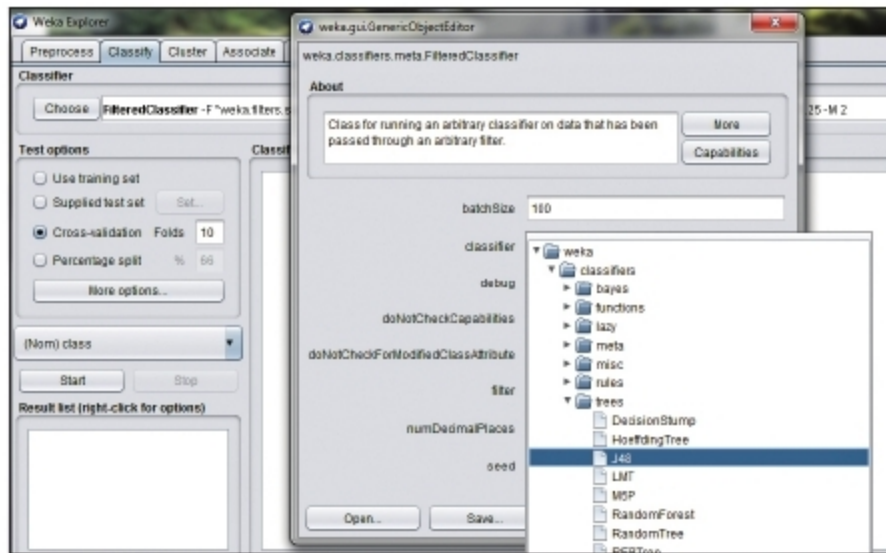


Figure 5: Selection of the classification algorithm

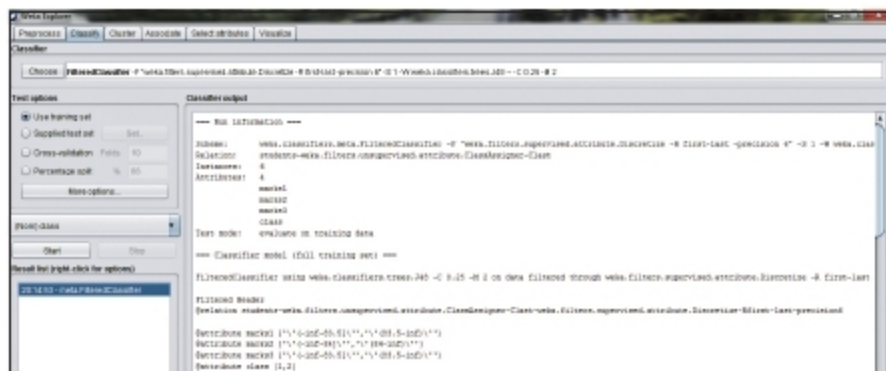


Figure 6: Running the classifier on the training data set

Figure 2 depicts the Weka GUI Chooser interface in which there are multiple options to work with the data science and machine learning based implementations. In this option, there are multiple options including Explorer, Experimenter, Knowledge Flow, Workbench and Simple CI. For traditional implementations, Explorer is used by data scientists, as it has user friendly interfaces to choose the data set and apply different algorithms without cramming any instructions or syntax for the algorithmic implementation.

From the Preprocess Tab in Weka, as shown in Figure 3, the training data can be selected. In the option to open the file, the data scientist can select the data set that is required to be trained for the modelling and processing of the classifier, as per the current scenario.

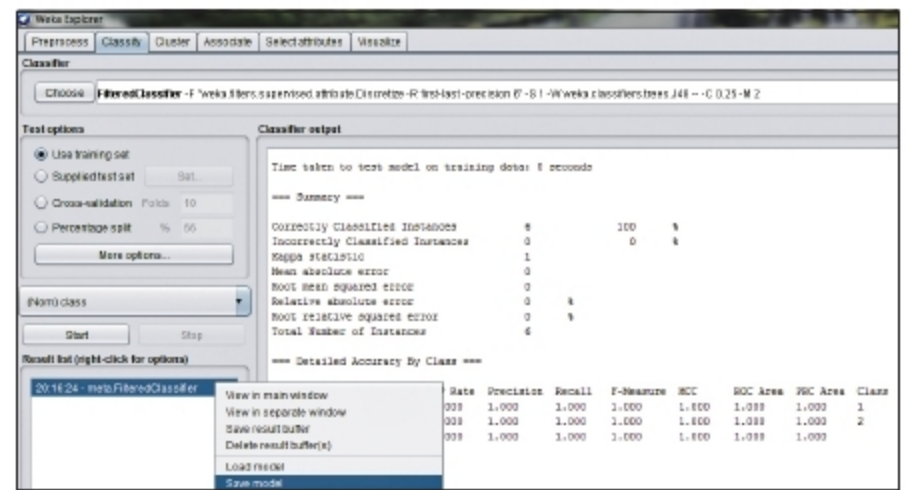


Figure 7: Saving the model for the prediction of test data

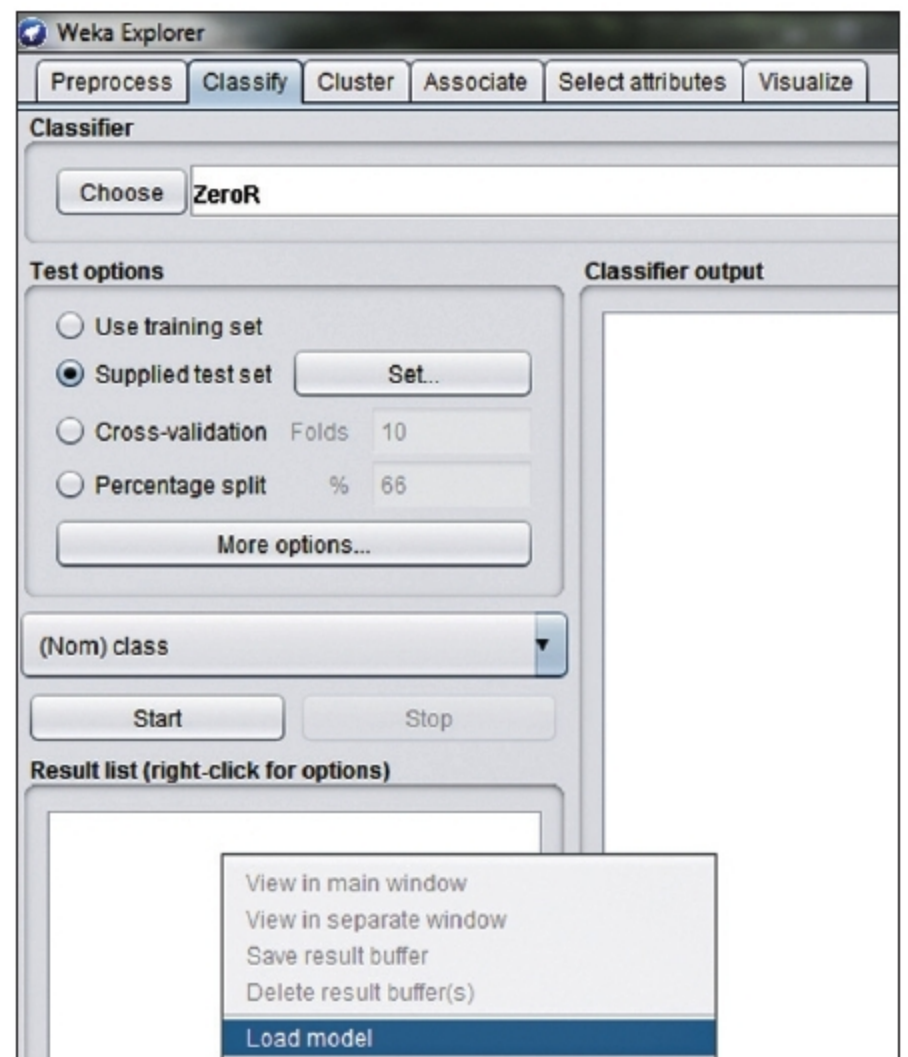


Figure 8: Loading the model for the evaluation and prediction of test data

Once the training data set is selected and imported to the Weka interface, the target class is required to be mentioned. As per the training data, the attribute 'class' is used here as the target. It means that the 'class' is the determined value on the combinations and associations of other attributes — marks1, marks2 and marks3.

In Weka, there are assorted algorithms for data science and machine learning that can be called and attached with the data set to be processed. Figure 5 presents the option to select the classification algorithm of J48 so that the classification model can be built on the basis of the training data selected earlier.

Once the data set and classification model are invoked, there is a need to run the classifier so that the model can be trained. In this way, the classification model is correlated with the association of determining attributes and the determined attribute. In this example of students, the determining